

Lecture 21: Positive Definite Matrices

Quadratic Forms, Ellipses, and the Question $A = M^T M$

The Question

Today's Question

Last time, one matrix appeared again and again:

$$M^T M.$$

For every vector x :

$$x^T M^T M x = (Mx)^T (Mx) = \|Mx\|^2.$$

So $M^T M$ always produces a **squared length**.

**Which symmetric matrices can be written as
 $M^T M$?**

The Necessary Condition

If

$$A = M^T M,$$

then:

$$x^T A x = x^T M^T M x = \|Mx\|^2.$$

Therefore:

$$x^T A x \geq 0 \quad \text{for every } x.$$

So a candidate A must never assign negative squared length.

A Symmetric Matrix Measures Length?

A real symmetric matrix A gives a quadratic expression:

$$q_A(x) = x^T A x.$$

If $q_A(x)$ behaves like squared length, then:

$$q_A(x) > 0 \quad \text{for every } x \neq 0.$$

If $q_A(x)$ sometimes becomes negative, it cannot be a squared length.

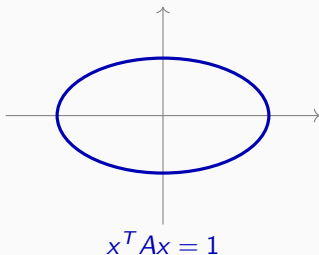
The whole lecture is about testing this sign.

Look at the Unit Shape

Instead of testing every vector one by one, draw the set:

$$\{x : x^T A x = 1\}.$$

This is the “unit circle” for the quadratic form.

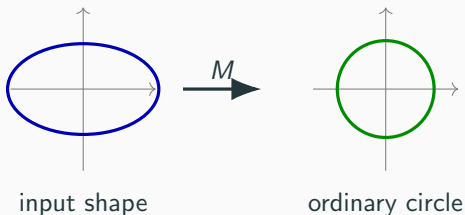


If It Comes From $M^T M$

If $A = M^T M$, then:

$$x^T A x = 1 \iff \|Mx\|^2 = 1.$$

So the shape is a preimage of an ordinary circle/sphere.

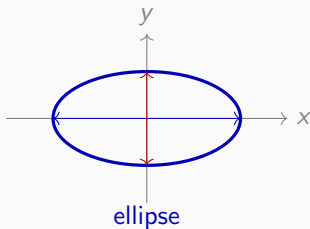


This suggests ellipse-type geometry, not hyperbola geometry.

Three Possible Shapes

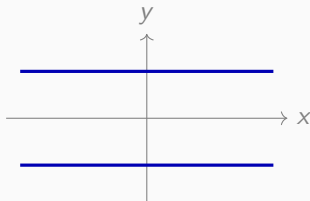
Good Case: Ellipse

$$A = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix}, \quad x^T A x = 1 \iff \frac{x^2}{4} + y^2 = 1.$$



Borderline Case: Two Lines

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad x^T A x = 1 \iff y^2 = 1.$$

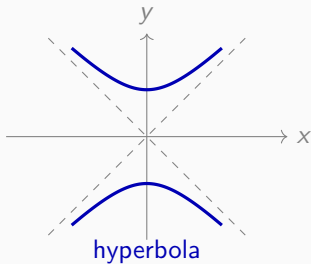


degenerate shape

One direction has zero length.

Bad Case: Hyperbola

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad x^T A x = 1 \iff -x^2 + y^2 = 1.$$



Three Shapes, Three Behaviors

Matrix	Shape of $x^T A x = 1$	Behavior
$\begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{pmatrix}$	ellipse	always positive
$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$	two lines	never negative, but degenerate
$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	hyperbola	sometimes negative

Now we need names for these three behaviors.

Definition

Let A be a real symmetric matrix.

- A is **positive definite** if

$$x^T A x > 0 \quad \text{for every } x \neq 0.$$

- A is **positive semidefinite** if

$$x^T A x \geq 0 \quad \text{for every } x.$$

- A is **indefinite** if $x^T A x$ takes both positive and negative values.

Matrix as an Inner Product Table

Where Does $x^T A x$ Come From?

The ordinary dot product is:

$$\langle v, w \rangle = v^T w.$$

A symmetric matrix A gives a new bilinear form:

$$\langle v, w \rangle_A = v^T A w.$$

Then the quadratic form is the self-pairing:

$$\langle x, x \rangle_A = x^T A x.$$

So A is trying to define a new geometry.

Symmetry Means $A = A^T$

For an inner-product-like geometry, we want:

$$\langle v, w \rangle_A = \langle w, v \rangle_A.$$

Compute:

$$\langle v, w \rangle_A = v^T A w.$$

Also:

$$\langle w, v \rangle_A = w^T A v = (w^T A v)^T = v^T A^T w.$$

So symmetry of the form is exactly:

$$A = A^T.$$

Matrix Entries Are Inner Products

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_A = \underbrace{\begin{pmatrix} \langle e_1, e_1 \rangle_A & \langle e_1, e_2 \rangle_A & \langle e_1, e_3 \rangle_A \\ \langle e_2, e_1 \rangle_A & \langle e_2, e_2 \rangle_A & \langle e_2, e_3 \rangle_A \\ \langle e_3, e_1 \rangle_A & \langle e_3, e_2 \rangle_A & \langle e_3, e_3 \rangle_A \end{pmatrix}}_{\text{inner product table of basis vectors}}.$$

$$\underbrace{a_{ij}}_{\text{matrix entry}} = \underbrace{e_i^T A e_j}_{\langle e_i, e_j \rangle_A}.$$

The matrix A is the inner product table of the standard basis.

Standard Basis Vectors

In $\mathbb{R}^3$:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ \textcolor{red}{1} \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The vector e_2 means:

select the second coordinate.

So multiplying by e_2 should select the second column or second row.

Right Multiplication Selects a Column

Let

$$A = \begin{pmatrix} a_{11} & \mathbf{a_{12}} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \\ a_{31} & \mathbf{a_{32}} & a_{33} \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ \mathbf{1} \\ 0 \end{pmatrix}.$$

Then:

$$Ae_2 = \begin{pmatrix} \mathbf{a_{12}} \\ \mathbf{a_{22}} \\ \mathbf{a_{32}} \end{pmatrix}.$$

Ae_2 selects column 2.

Left Multiplication Selects a Row

Now put e_2^T on the left:

$$e_2^T = \begin{pmatrix} 0 & \mathbf{1} & 0 \end{pmatrix}.$$

Then:

$$e_2^T A = \begin{pmatrix} \mathbf{a_{21}} & \mathbf{a_{22}} & \mathbf{a_{23}} \end{pmatrix}.$$

$e_2^T A$ selects row 2.

Diagonal Entry = Pivot Value

Put both together:

$$e_2^T A e_2 = \begin{pmatrix} 0 & \mathbf{1} & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 \\ \mathbf{1} \\ 0 \end{pmatrix}.$$

Result:

$$e_2^T A e_2 = \mathbf{a_{22}}.$$

This is the diagonal pivot used later in diagonal cross-filling.

Basis Table Picture

The whole matrix is a table:

$$A = \begin{pmatrix} \langle e_1, e_1 \rangle_A & \langle e_1, e_2 \rangle_A & \langle e_1, e_3 \rangle_A \\ \langle e_2, e_1 \rangle_A & \langle e_2, e_2 \rangle_A & \langle e_2, e_3 \rangle_A \\ \langle e_3, e_1 \rangle_A & \langle e_3, e_2 \rangle_A & \langle e_3, e_3 \rangle_A \end{pmatrix}.$$

The diagonal entries are self-lengths:

$$a_{ii} = e_i^T A e_i = \langle e_i, e_i \rangle_A.$$

Eigenvalue Criterion

A Fast Positive Example

Consider

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

Its eigenvectors are:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The eigenvalues are:

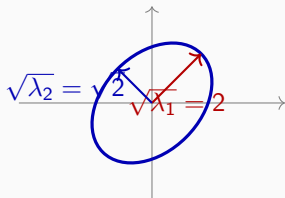
$$\lambda_1 = 4, \quad \lambda_2 = 2.$$

Both eigenvalues are positive: ellipse geometry.

Ellipse from Positive Eigenvalues

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\lambda_1 = 4, \quad \lambda_2 = 2.$$



$$\underbrace{\sqrt{\lambda_1}}_{\text{semi-axis length}} = 2, \quad \underbrace{\sqrt{\lambda_2}}_{\text{semi-axis length}} = \sqrt{2}.$$

Eigenvalues are squared coefficients; semi-axis lengths use square roots.

A Fast Negative Example

Now consider

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

The same two directions appear:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

But the eigenvalues are:

$$\lambda_1 = 3, \quad \lambda_2 = -1.$$

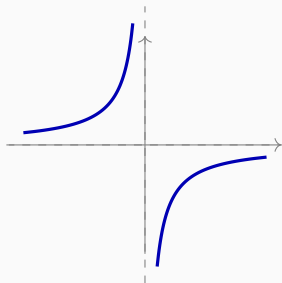
One negative eigenvalue: not squared length.

Hyperbola from a Negative Eigenvalue

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\lambda_1 = 3, \quad \lambda_2 = -1.$$

One direction contributes negative squared length.



A negative eigenvalue produces hyperbola-type geometry.

Eigenvalue Criterion

Theorem (Eigenvalue Criterion)

Let A be a real symmetric matrix.

- A is positive definite iff all eigenvalues are positive.
- A is positive semidefinite iff all eigenvalues are nonnegative.
- A is indefinite if it has both positive and negative eigenvalues.

This is the fastest conceptual test.

Why This Criterion Works

From the real symmetric spectral theorem:

$$A = \Omega \Lambda \Omega^T, \quad \Omega^T \Omega = I.$$

Then:

$$x^T A x = x^T \Omega \Lambda \Omega^T x.$$

Set:

$$y = \Omega^T x.$$

So:

$$x^T A x = y^T \Lambda y.$$

Diagonal Form Shows the Signs

If

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \end{pmatrix},$$

then:

$$y^T \Lambda y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots .$$

The signs of the eigenvalues control the signs of $x^T A x$.

Eigenvalue Criterion Summary

Eigenvalues	Form	Shape
all > 0	positive except 0	ellipse
all ≥ 0	never negative	degenerate
mixed signs	both signs	hyperbola

The spectral theorem turns positivity into a diagonal sign check.

Spectral Factorization

Return to the Opening Question

We asked:

$$\text{When can } A = M^T M?$$

We already know one direction:

$$A = M^T M \implies x^T A x = \|Mx\|^2 \geq 0.$$

So:

$$A = M^T M \implies A \text{ is positive semidefinite.}$$

The Converse Should Use Eigenvalues

Suppose A is positive semidefinite.

By the eigenvalue criterion:

$$A = \Omega \Lambda \Omega^T, \quad \lambda_i \geq 0.$$

Since the eigenvalues are nonnegative, we can take square roots:

$$\sqrt{\Lambda} = \begin{pmatrix} \sqrt{\lambda_1} & & \\ & \sqrt{\lambda_2} & \\ & & \ddots \end{pmatrix}.$$

This is exactly where positive semidefinite matters.

Build the Factor

Since

$$\Lambda = \sqrt{\Lambda}\sqrt{\Lambda},$$

we get:

$$A = \Omega\sqrt{\Lambda}\sqrt{\Lambda}\Omega^T.$$

Now regroup:

$$A = (\sqrt{\Lambda}\Omega^T)^T(\sqrt{\Lambda}\Omega^T).$$

Therefore take:

$$\boxed{M = \sqrt{\Lambda}\Omega^T.}$$

Spectral Factorization Criterion

Theorem (Spectral Factorization Criterion)

For a real symmetric matrix A ,

$$A = M^T M \text{ for some } M$$

if and only if

A is positive semidefinite.

If M has linearly independent columns, then:

$$x^T M^T M x = \|Mx\|^2 > 0 \quad (x \neq 0),$$

so $M^T M$ is positive definite.

There Will Be a Second Factorization

$$\underbrace{A = \Omega \Lambda \Omega^T}_{\text{spectral route}} \implies \underbrace{A = M^T M}_{\text{factorization by eigenvalues}} .$$

$$\underbrace{A = P_1 + P_2 + \cdots + P_r}_{\text{diagonal cross-filling route}} \implies \underbrace{A = NN^T}_{\text{factorization by cross-filled columns}} .$$

So the eigenvalue construction is one factorization, not the end of the story.

Concrete Positive Example

For

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix},$$

we have eigenvalues 4 and 2.

So:

$$\sqrt{\Lambda} = \begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix}.$$

With orthogonal eigenvector matrix Ω :

$$A = (\sqrt{\Lambda}\Omega^T)^T(\sqrt{\Lambda}\Omega^T).$$

What We Have So Far

We now have two criteria:

Criterion	Test
Eigenvalues	all $\lambda_i \geq 0$ or > 0
Factorization	$A = M^T M$

Next: a cross-filling criterion that does not start by finding eigenvalues.

Checking Semipositivity by Diagonal Cross-Filling

Cross-Filling Criterion: Two Directions

$\underbrace{\text{positive-pivot cross-filling succeeds}}_{\substack{\text{every used pivot } p_k > 0 \\ \text{zero rows/columns may be skipped}}} \implies \underbrace{A = MM^T}_{\text{stack cross-filled columns}} \implies \underbrace{A \succeq 0}_{\text{easy direction}} .$

$\underbrace{A \succeq 0}_{\text{hard direction}} \implies \underbrace{\text{cross-filling can proceed legally}}_{\text{needs three modules}} .$

The easy direction constructs MM^T .
 The hard direction proves PSD matrices never get stuck.

Easy Direction: Positive Pivots Build Columns

$$\underbrace{\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}}_{\text{pivot column}} \quad \underbrace{b_2 > 0}_{\text{positive pivot}} \quad \Rightarrow \quad m = \underbrace{\frac{1}{\sqrt{b_2}}}_{\text{cross-filled column}} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

$$\underbrace{mm^T}_{\text{one PSD piece}} = \frac{1}{b_2} \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} (b_1 \quad b_2 \quad b_3)}_P.$$

The positive pivot is required because the column uses $\sqrt{b_2}$.

Easy Direction: Stack All Cross-Filled Columns

$$\underbrace{A}_{\text{after successful cross-filling}} = \underbrace{m_1 m_1^T + m_2 m_2^T + \cdots + m_r m_r^T}_{\text{rank-one PSD pieces}}.$$

$$\underbrace{M = \begin{pmatrix} | & | & & | \\ m_1 & m_2 & \cdots & m_r \\ | & | & & | \end{pmatrix}}_{\text{stack the cross-filled columns}} \implies \underbrace{A = MM^T}_{\text{factorization}}.$$

$$\underbrace{x^T A x}_{\text{test vector}} = \underbrace{x^T M M^T x}_{\text{factorized form}} = \underbrace{\|M^T x\|^2}_{\geq 0}.$$

Hard Direction: Three Modules

$$\underbrace{A \succeq 0}_{\text{input assumption}} \implies \left\{ \begin{array}{l} \underbrace{a_{ii} = e_i^T A e_i \geq 0}_{\text{Module 1: no negative diagonal}}, \\ \underbrace{a_{ii} = 0 \Rightarrow \text{row } i \text{ and column } i \text{ are zero}}_{\text{Module 2: zero pivot is harmless}}, \\ \underbrace{a_{ii} > 0 \Rightarrow A = P + R, \quad P \succeq 0, \quad R \succeq 0}_{\text{Module 3: positive pivot keeps induction alive}}. \end{array} \right.$$

zero pivot $\Rightarrow$ skip	positive pivot $\Rightarrow$ cross-fill and continue.
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One Diagonal Cross-Fill Step

Use the second diagonal entry as the visible pivot:

$$a_{22} = e_2^T A e_2 > 0.$$

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_A = \underbrace{\begin{pmatrix} p_{11} & a_{12} & p_{13} \\ a_{21} & a_{22} & a_{23} \\ p_{31} & a_{32} & p_{33} \end{pmatrix}}_P + \underbrace{\begin{pmatrix} r_{11} & 0 & r_{13} \\ 0 & 0 & 0 \\ r_{31} & 0 & r_{33} \end{pmatrix}}_{R=A-P}.$$

Diagonal cross-filling means: honestly write the two matrices P and R .

Entries of the Rank-One Piece

At the pivot a_{22} , write the pivot column entries as

$$b_1 = a_{12}, \quad b_2 = a_{22}, \quad b_3 = a_{32}.$$

$$\underbrace{\begin{pmatrix} \frac{b_1 b_1}{b_2} & b_1 & \frac{b_1 b_3}{b_2} \\ b_1 & \textcolor{red}{b_2} & b_3 \\ \frac{b_3 b_1}{b_2} & b_3 & \frac{b_3 b_3}{b_2} \end{pmatrix}}_P$$

This is the honest entrywise matrix display of the rank-one piece.

Inner Product Meaning of the Entries

$$\underbrace{\begin{pmatrix} \frac{\langle \textcolor{blue}{e}_1, \textcolor{red}{e}_2 \rangle_A \langle \textcolor{red}{e}_2, \textcolor{blue}{e}_1 \rangle_A}{\langle \textcolor{red}{e}_2, \textcolor{red}{e}_2 \rangle_A} & \langle \textcolor{blue}{e}_1, \textcolor{red}{e}_2 \rangle_A & \frac{\langle \textcolor{blue}{e}_1, \textcolor{red}{e}_2 \rangle_A \langle \textcolor{red}{e}_2, \textcolor{green}{e}_3 \rangle_A}{\langle \textcolor{red}{e}_2, \textcolor{red}{e}_2 \rangle_A} \\ \langle \textcolor{red}{e}_2, \textcolor{blue}{e}_1 \rangle_A & \langle \textcolor{red}{e}_2, \textcolor{red}{e}_2 \rangle_A & \langle \textcolor{red}{e}_2, \textcolor{green}{e}_3 \rangle_A \\ \frac{\langle \textcolor{green}{e}_3, \textcolor{red}{e}_2 \rangle_A \langle \textcolor{red}{e}_2, \textcolor{blue}{e}_1 \rangle_A}{\langle \textcolor{red}{e}_2, \textcolor{red}{e}_2 \rangle_A} & \langle \textcolor{green}{e}_3, \textcolor{red}{e}_2 \rangle_A & \frac{\langle \textcolor{green}{e}_3, \textcolor{red}{e}_2 \rangle_A \langle \textcolor{red}{e}_2, \textcolor{green}{e}_3 \rangle_A}{\langle \textcolor{red}{e}_2, \textcolor{red}{e}_2 \rangle_A} \end{pmatrix}}_{P \text{ written entrywise from colored basis vectors}}$$

$\underbrace{\textcolor{blue}{e}_1}_{\text{first row/column}} \quad \underbrace{\textcolor{red}{e}_2}_{\text{pivot direction}} \quad \underbrace{\textcolor{green}{e}_3}_{\text{third row/column}} .$

Only the vectors are colored; each inner product remains one object.

Module 1: No Negative Diagonal

$$\underbrace{A \succeq 0}_{\text{PSD assumption}} \implies \underbrace{a_{ii} = e_i^T A e_i \geq 0}_{\text{test vector } e_i} .$$

$\underbrace{a_{ii} < 0}_{\text{negative diagonal appears}} \implies \underbrace{A \not\succeq 0}_{\text{stop cross-filling test}} .$

This explains why negative pivots immediately fail the PSD test.

Lemma: Cauchy Is the Tool for Continuation

$$\underbrace{A \succeq 0}_{\text{PSD form}} \implies \underbrace{\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A}_{\text{Cauchy lemma}}.$$

Proposition (Cauchy Inequality for PSD Forms)

If A is positive semidefinite and $\langle x, y \rangle_A = x^T A y$, then

$$\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.$$

Use 1 : $a_{ii} = 0 \Rightarrow$ zero row/column,

Use 2 : $a_{ii} > 0 \Rightarrow R = A - P \succeq 0$.

The purpose of Cauchy is to keep cross-filling alive.

Module 2: Zero Diagonal Means Zero Row

Proposition (Zero Diagonal Lemma)

If $A \succeq 0$ and

$$a_{ii} = e_i^T A e_i = 0,$$

then the whole i -th row and the whole i -th column are zero.

For any j , Cauchy gives:

$$a_{ij}^2 = \langle e_i, e_j \rangle_A^2 \leq \langle e_i, e_i \rangle_A \langle e_j, e_j \rangle_A = 0 \cdot a_{jj} = 0.$$

So $a_{ij} = 0$ for every j .

Module 2 Meaning: Skip Zero Rows

If $A \succeq 0$ and a diagonal entry is zero:

$$a_{ii} = 0,$$

then the whole row and column are already zero.

$$\begin{pmatrix} * & * & 0 & * \\ * & * & 0 & * \\ 0 & 0 & 0 & 0 \\ * & * & 0 & * \end{pmatrix}$$

There is no need to choose this as a pivot. Skip it.

Module 3: Positive Pivot Splits PSD

Now assume the diagonal pivot is positive:

$$a_{ii} = e_i^T A e_i > 0.$$

Proposition (Positive Pivot Step)

If $A \succeq 0$ and $a_{ii} > 0$, then diagonal cross-filling gives

$$A = P + R, \quad P \succeq 0, \quad R \succeq 0.$$

Now prove it by looking at the actual entries of P and R .

Module 3 Setup: Write A as Inner Products

$$\underbrace{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A}_{\text{positive pivot}} > 0$$

$$\underbrace{\begin{pmatrix} \langle \mathbf{e}_1, \mathbf{e}_1 \rangle_A & \langle \mathbf{e}_1, \mathbf{e}_2 \rangle_A & \langle \mathbf{e}_1, \mathbf{e}_3 \rangle_A \\ \langle \mathbf{e}_2, \mathbf{e}_1 \rangle_A & \langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A & \langle \mathbf{e}_2, \mathbf{e}_3 \rangle_A \\ \langle \mathbf{e}_3, \mathbf{e}_1 \rangle_A & \langle \mathbf{e}_3, \mathbf{e}_2 \rangle_A & \langle \mathbf{e}_3, \mathbf{e}_3 \rangle_A \end{pmatrix}}_{A \text{ as an inner product table}}$$

Only the basis vectors are colored. Each inner product stays one object.

Module 3: The Cross-Filled Piece P

$$\underbrace{\begin{pmatrix} \frac{\langle \mathbf{e}_1, \mathbf{e}_2 \rangle_A \langle \mathbf{e}_2, \mathbf{e}_1 \rangle_A}{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A} & \langle \mathbf{e}_1, \mathbf{e}_2 \rangle_A & \frac{\langle \mathbf{e}_1, \mathbf{e}_2 \rangle_A \langle \mathbf{e}_2, \mathbf{e}_3 \rangle_A}{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A} \\ \langle \mathbf{e}_2, \mathbf{e}_1 \rangle_A & \langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A & \langle \mathbf{e}_2, \mathbf{e}_3 \rangle_A \\ \frac{\langle \mathbf{e}_3, \mathbf{e}_2 \rangle_A \langle \mathbf{e}_2, \mathbf{e}_1 \rangle_A}{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A} & \langle \mathbf{e}_3, \mathbf{e}_2 \rangle_A & \frac{\langle \mathbf{e}_3, \mathbf{e}_2 \rangle_A \langle \mathbf{e}_2, \mathbf{e}_3 \rangle_A}{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A} \end{pmatrix}}_P$$

$$\underbrace{P = mm^T}_{\text{PSD piece}}, \quad m = \frac{1}{\underbrace{\sqrt{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A}}_{\text{positive pivot gives the square root}}} \begin{pmatrix} \langle \mathbf{e}_1, \mathbf{e}_2 \rangle_A \\ \langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A \\ \langle \mathbf{e}_3, \mathbf{e}_2 \rangle_A \end{pmatrix}.$$

Module 3: The Remainder $R = A - P$

$$\underbrace{A-P}^R = \underbrace{\begin{pmatrix} \langle e_1, e_1 \rangle_A - \frac{\langle e_1, e_2 \rangle_A \langle e_2, e_1 \rangle_A}{\langle e_2, e_2 \rangle_A} & 0 & \langle e_1, e_3 \rangle_A - \frac{\langle e_1, e_2 \rangle_A \langle e_2, e_3 \rangle_A}{\langle e_2, e_2 \rangle_A} \\ 0 & 0 & 0 \\ \langle e_3, e_1 \rangle_A - \frac{\langle e_3, e_2 \rangle_A \langle e_2, e_1 \rangle_A}{\langle e_2, e_2 \rangle_A} & 0 & \langle e_3, e_3 \rangle_A - \frac{\langle e_3, e_2 \rangle_A \langle e_2, e_3 \rangle_A}{\langle e_2, e_2 \rangle_A} \end{pmatrix}}_{\text{entrywise remainder}}$$

The pivot row and pivot column are zero; only the remaining block must stay PSD.

Module 3: Test the Remainder

$$\underbrace{v}_{\text{testing vector}} = \underbrace{x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3}_{\text{any vector}}.$$

$$\underbrace{v^T R v}_{\text{test } R \text{ on both sides}} = \underbrace{v^T (A - P) v}_{R=A-P} = \underbrace{\langle v, v \rangle_A}_{v^T A v} - \underbrace{\frac{\langle v, \mathbf{e}_2 \rangle_A^2}{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A}}_{v^T P v}.$$

$$\underbrace{\langle v, \mathbf{e}_2 \rangle_A^2}_{\text{Cauchy numerator}} \leq \underbrace{\langle v, v \rangle_A \langle \mathbf{e}_2, \mathbf{e}_2 \rangle_A}_{\text{Cauchy denominator clears}} \implies \boxed{v^T R v \geq 0}.$$

Cauchy is exactly the reason the remainder remains positive semidefinite.

Cross-Filling Criterion So Far

For $A \succeq 0$:

Diagonal pivot	What happens
$a_{ii} < 0$	impossible
$a_{ii} = 0$	row/column must be zero; skip it
$a_{ii} > 0$	cross-fill: $A = P + (A - P)$ with both PSD

This is the diagonal cross-filling behavior we wanted.

Proof of Cauchy: Geometric Memory

Proposition (Cauchy Inequality for PSD Forms)

If $A \succeq 0$ and $\langle x, y \rangle_A = x^T A y$, then

$$\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.$$

$$\underbrace{A = I}_{\text{ordinary dot product}} \implies \underbrace{(x^T y)^2 \leq (x^T x)(y^T y)}_{\text{circle intuition only}}.$$

This page is only for geometric memory, not the proof.

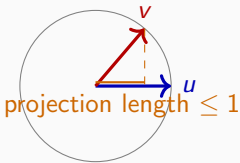
Cauchy: Picture

For the ordinary dot product:

$$(x^T y)^2 \leq (x^T x)(y^T y).$$

After normalizing:

$$\left(\frac{x}{\|x\|} \right)^T \left(\frac{y}{\|y\|} \right) \leq 1.$$



Cauchy Proof: Two Cases

Proposition (Cauchy Inequality for PSD Forms)

If $A \succeq 0$, then

$$\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.$$

Case	Proof method
$\langle x, x \rangle_A > 0$	build a cancellation vector w
$\langle x, x \rangle_A = 0$	use the polynomial in t

If $\langle z, z \rangle_A = 0$ for every z , then $A = 0$ and Cauchy is trivial.

So the proof is really Case 1 plus Case 2.

Case 1: Positive Length

Assume the first case:

$$\langle x, x \rangle_A > 0.$$

Build a vector designed to cancel the x -component of y :

$$w = \langle x, x \rangle_A y - \langle x, y \rangle_A x.$$

Since A is positive semidefinite:

$$0 \leq \langle w, w \rangle_A.$$

Now expand this one inequality.

Expand $\langle w, w \rangle_A$

Let

$$\alpha = \langle x, x \rangle_A, \quad \beta = \langle x, y \rangle_A.$$

Then:

$$w = \alpha y - \beta x.$$

So:

$$0 \leq \langle w, w \rangle_A = \langle \alpha y - \beta x, \alpha y - \beta x \rangle_A.$$

Using bilinearity:

$$0 \leq \alpha^2 \langle y, y \rangle_A - 2\alpha\beta \langle x, y \rangle_A + \beta^2 \langle x, x \rangle_A.$$

Finish Cauchy

Since

$$\beta = \langle x, y \rangle_A, \quad \alpha = \langle x, x \rangle_A,$$

the expansion becomes:

$$0 \leq \alpha^2 \langle y, y \rangle_A - 2\alpha\beta^2 + \beta^2\alpha.$$

Combine the last two terms:

$$0 \leq \alpha^2 \langle y, y \rangle_A - \alpha\beta^2.$$

Finish Cauchy (continued)

We have:

$$0 \leq \alpha^2 \langle y, y \rangle_A - \alpha \beta^2.$$

Divide by $\alpha = \langle x, x \rangle_A > 0$:

$$0 \leq \langle x, x \rangle_A \langle y, y \rangle_A - \langle x, y \rangle_A^2.$$

Therefore:

$$\boxed{\langle x, y \rangle_A^2 \leq \langle x, x \rangle_A \langle y, y \rangle_A.}$$

Case 2: Zero Length

What if $\langle x, x \rangle_A = 0$?

For any real number t :

$$0 \leq \langle x + ty, x + ty \rangle_A.$$

Expanding:

$$0 \leq 2t\langle x, y \rangle_A + t^2\langle y, y \rangle_A.$$

This quadratic cannot be nonnegative for all t unless:

$\langle x, y \rangle_A = 0.$

Worked Cross-Filling Tests

Test 1: Zero Diagonal Obstruction

Consider

$$\underbrace{\begin{pmatrix} 1 & 2 & -3 \\ 2 & 0 & 2 \\ -3 & 2 & 5 \end{pmatrix}}_A$$

If $A \succeq 0$ and $a_{22} = 0$, the zero diagonal lemma forces the whole second row and column to be zero.

But here:

$$a_{21} = 2, \quad a_{23} = 2.$$

Therefore A is not positive semidefinite.

Test 2: Negative Remainder Obstruction

Start with

$$\underbrace{\begin{pmatrix} \color{red}{1} & \color{blue}{3} & \color{blue}{2} \\ \color{blue}{3} & 5 & 4 \\ \color{blue}{2} & 4 & 9 \end{pmatrix}}_B.$$

Cross-fill from the first diagonal pivot:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 5 & 4 \\ 2 & 4 & 9 \end{pmatrix}}_B - \underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 9 & 6 \\ 2 & 6 & 4 \end{pmatrix}}_P = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \color{red}{-4} & -2 \\ 0 & -2 & 5 \end{pmatrix}}_R.$$

A PSD remainder cannot have a negative diagonal entry.

Test 3: Positive Cross-Filling

Now consider

$$\underbrace{\begin{pmatrix} \textcolor{red}{1} & \textcolor{blue}{3} & \textcolor{blue}{2} \\ \textcolor{blue}{3} & 10 & 7 \\ \textcolor{blue}{2} & 7 & 9 \end{pmatrix}}_A.$$

First diagonal cross-fill:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_A = \underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 9 & 6 \\ 2 & 6 & 4 \end{pmatrix}}_{P_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{pmatrix}}_{R_1}.$$

Test 3: Continue

The active remainder is the lower-right block:

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \color{red}{1} & \color{blue}{1} \\ 0 & \color{blue}{1} & 5 \end{pmatrix}}_{R_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}}_{P_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \color{red}{4} \end{pmatrix}}_{R_2}.$$

Final pivot:

$$R_2 = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}}_{P_3}.$$

Test 3: Sum of Positive Pieces

The whole matrix is a sum of positive rank-one pieces:

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_A = \underbrace{\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} (1 \ 3 \ 2)}_{P_1} + \underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} (0 \ 1 \ 1)}_{P_2} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} (0 \ 0 \ 2)}_{P_3}.$$

Therefore:

$$x^T A x = (x_1 + 3x_2 + 2x_3)^2 + (x_2 + x_3)^2 + (2x_3)^2$$

This is > 0 for every $x \neq 0$.

Test 3 Also Builds a Factorization

$$\underbrace{\begin{pmatrix} 1 & 3 & 2 \\ 3 & 10 & 7 \\ 2 & 7 & 9 \end{pmatrix}}_A = \underbrace{\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}}_{m_1 m_1^T} + \underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}}_{m_2 m_2^T} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 \end{pmatrix}}_{m_3 m_3^T}.$$

$$\underbrace{M = \begin{pmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{pmatrix}}_{\text{stack the cross-filled columns}} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 2 \end{pmatrix} \implies \boxed{A = MM^T}.$$

Cross-filling is both a test and a constructive factorization.

Cross-Filling Criterion

For a real symmetric matrix:

Cross-filling behavior	Conclusion
positive pivots until zero remainder	positive definite
positive pivots + zero rows/columns	positive semidefinite
zero diagonal but nonzero row	not PSD
negative diagonal in remainder	not PSD

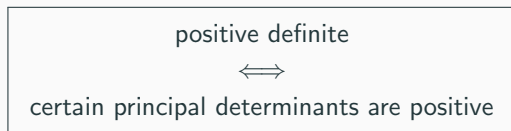
This is the computational test from diagonal cross-filling.

Connection to Determinants

What the Literature Says

Many textbooks do not say “cross-filling.”

They usually say:



For example, in the no-row-swap diagonal order:

$$\det A_1 > 0, \quad \det A_2 > 0, \quad \cdots, \quad \det A_n > 0.$$

This is Sylvester's determinant criterion.

Principal Determinants

For a 3×3 matrix, leading principal determinants are:

$$\underbrace{\begin{pmatrix} a_{11} & * & * \\ * & * & * \\ * & * & * \end{pmatrix}}_{A_1} \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & * \\ a_{21} & a_{22} & * \\ * & * & * \end{pmatrix}}_{A_2} \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}_{A_3=A}.$$

The literature checks:

$$\det A_1, \quad \det A_2, \quad \det A_3.$$

Our Dictionary: Determinant = Product of Pivots

Diagonal cross-filling produces pivots:

$$d_1, d_2, \dots, d_n.$$

From our determinant lecture:

$$\det A_k = d_1 d_2 \cdots d_k.$$

So the next pivot is the ratio:

$$d_k = \frac{\det A_k}{\det A_{k-1}}.$$

Principal determinants are just cumulative products of cross-filling pivots.

Positive Definite: Same Test, Different Language

Cross-filling says:

$$d_1 > 0, \quad d_2 > 0, \quad \dots, \quad d_n > 0.$$

Determinants say:

$$\det A_1 > 0, \quad \det A_2 > 0, \quad \dots, \quad \det A_n > 0.$$

They are equivalent because:

$$\det A_k = d_1 d_2 \cdots d_k.$$

Same information: determinant criterion hides the pivots inside products.

Why This Breaks for Semidefinite

Leading determinants only see products:

$$d_1 = 0 \implies \det A_1 = \det A_2 = \dots = 0$$

even if some hidden direction is negative.

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_B$$

Here B is not PSD, but:

$$\det B_1 = 0, \quad \det B_2 = 0, \quad \det B_3 = 0.$$

Leading determinant tests cannot certify semidefinite matrices.

Semidefinite: Why Literature Becomes Heavier

For positive semidefinite matrices, zero pivots can appear.

Then leading determinants alone are not enough. Textbooks usually say:

all principal minors must be nonnegative.

That means many determinants, even in 3×3 :

$$\begin{array}{ccc} \det A_{\{1\}}, & \det A_{\{2\}}, & \det A_{\{3\}}, \\ \det A_{\{1,2\}}, & \det A_{\{1,3\}}, & \det A_{\{2,3\}}, \\ & \det A_{\{1,2,3\}}. \end{array}$$

Our zero-row rule replaces this determinant explosion with structure.

Complexity: Determinants vs. Pivots

For an $n \times n$ matrix, all principal minors means one determinant for every nonempty subset:

$$2^n - 1 \text{ determinants.}$$

Cross-filling checks the active diagonal behavior directly:

$$\text{pivots } d_k \quad + \quad \text{zero row/column rule} \quad + \quad \text{remainders.}$$

Literature PSD test	Cross-filling PSD test
many principal determinants	pivots and remainders
products hide zero pivots	zero pivots are inspected
sign answer only	structural obstruction

Why Cross-Filling Is Stronger

Determinants answer only one question:

$\det(\text{submatrix})$ positive, zero, or negative?

Cross-filling gives more data:

Cross-filling data	Meaning
pivot d_k	determinant ratio
rank-one piece P_k	squared-length component
remainder R_k	next active problem
zero row/column	semidefinite degeneracy
negative diagonal in R_k	obstruction

Computing determinants is essentially cross-filling with less memory.

Applications

Application 1: Square Root of a PSD Matrix

$$\underbrace{A \succeq 0}_{\text{positive semidefinite}} \implies A = \underbrace{\Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T}_{\substack{\text{orthogonal diagonalization} \\ \lambda_j \geq 0}}.$$

$$A^{1/2} := \underbrace{\Omega \begin{pmatrix} \sqrt{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\lambda_n} \end{pmatrix} \Omega^T}_{\text{define one symmetric square root}}.$$

$$\underbrace{A^{1/2} A^{1/2}}_{\text{multiply in the same eigenbasis}} = \Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T = A.$$

Matrix square roots are not unique; this defines the symmetric PSD square root.

Application 1: Example

$$\underbrace{A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}}_{\text{PSD}} = \underbrace{\Omega}_{\text{orthogonal eigenbasis}} \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}}_{\Lambda} \underbrace{\Omega^T}_{\text{eigen-coordinate rows}} .$$

$$\underbrace{A^{1/2} := \Omega \begin{pmatrix} \sqrt{4} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \Omega^T}_{\text{replace squared axis coefficients by axis coefficients}} = \underbrace{\Omega \begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix} \Omega^T}_{\text{symmetric square root}} .$$

PSD matrices allow a real symmetric square root.

Application 2: Comparing Two Inner Products

$$A = \underbrace{\Omega \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Omega^T}_{\lambda_{\min} \leq \lambda_i \leq \lambda_{\max}} \quad \underbrace{y = \Omega^T x}_{\text{orthogonal change of coordinates}}.$$

$$\underbrace{x^T A x}_{\langle x, x \rangle_A} = y^T \underbrace{\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}}_{\lambda_1 y_1^2 + \dots + \lambda_n y_n^2} y.$$

$$\underbrace{\lambda_{\min}(y_1^2 + \dots + y_n^2)}_{\lambda_{\min} \|x\|^2} \leq \underbrace{x^T A x}_{\text{new squared length}} \leq \underbrace{\lambda_{\max}(y_1^2 + \dots + y_n^2)}_{\lambda_{\max} \|x\|^2}.$$

Eigenvalue Bound for the Ratio

$$\lambda_{\min} \leq \underbrace{\frac{x^T A x}{x^T x}}_{\text{ratio of new squared length to old squared length}} \leq \lambda_{\max} \quad (x \neq 0).$$

Theorem (Eigenvalue Comparison)

If A is real symmetric with smallest eigenvalue $\lambda_{\min}$ and largest eigenvalue $\lambda_{\max}$, then every nonzero vector x satisfies

$$\lambda_{\min} \leq \frac{x^T A x}{x^T x} \leq \lambda_{\max}.$$

Eigenvalues measure the strongest and weakest stretching of the inner product.

Final Summary

Complete answer:

$$A = M^T M \iff x^T A x \geq 0 \text{ for all } x.$$

Test	PSD criterion
quadratic form	$x^T A x \geq 0$ for all x
eigenvalues	every $\lambda_i \geq 0$
factorization	$A = M^T M$
cross-filling	positive pivots + zero-row rule

Positive definite means the same story with strict positivity.